Question 21: Bankers Algorithm

Source Code (C++):

```
#include <iostream>
using namespace std;
int main() {
    cout << "Name : Mohd Anas\n";
    cout << "Roll : 24BCS061\n";
    cout << "Bankers Algorithm\n\n";
    int n, m;
    cout << "Enter number of processes: ";
    cin >> n;
    cout << "Enter number of resource types: ";
    cin >> m;
    int alloc[n][m], max[n][m], need[n][m];
    int avail[m];
    cout << "\nEnter Allocation Matrix:\n";
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++)
            cin >> alloc[i][j];
    }
    cout << "\nEnter Max Matrix:\n";
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++)
            cin >> max[i][j];
    }
    cout << "\nEnter Available Resources:\n";
    for(int i = 0; i < m; i++)        cin >> avail[i];
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++)       need[i][j] = max[i][j]
- alloc[i][j];
    }
    bool finish[n] = {false};
    int safeSeq[n];
    int work[m];
    for(int i = 0; i < m; i++)        work[i] = avail[i];
    int count = 0;
    while(count < n) {
        bool found = false;
        for(int i = 0; i < n; i++) {
            if(!finish[i]) {
                bool canRun = true;
                for(int j = 0; j < m; j++) {
                    if(need[i][j] > work[j]) {
                        canRun = false;
```

```
                        break;
                    }
                }
                if(canRun) {
                    for(int j = 0; j < m;
j++)                  work[j] += alloc[i][j];
                   safeSeq[count++] = i;
                    finish[i] = true;
                    found = true;
                }
            }
        }
        if(!found) {
            cout << "\nSystem is NOT in a safe state.\n";
            return 0;
        }
    }
    cout << "\nSystem is in a SAFE state.\nSafe sequence: ";
    for(int i = 0; i < n; i++) {
        cout << "P" << safeSeq[i];
        if(i != n - 1) cout << " -> ";
    }
    cout << endl;
    return 0;
}
```

Output:

```
Name : Mohd Anas
Roll : 24BCS061
Bankers Algorithm

Enter number of processes: 5
Enter number of resource types: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

Enter Max Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3

Enter Available Resources:
3 3 2

System is in a SAFE state.
Safe sequence: P1 -> P3 -> P4 -> P0 -> P2
```